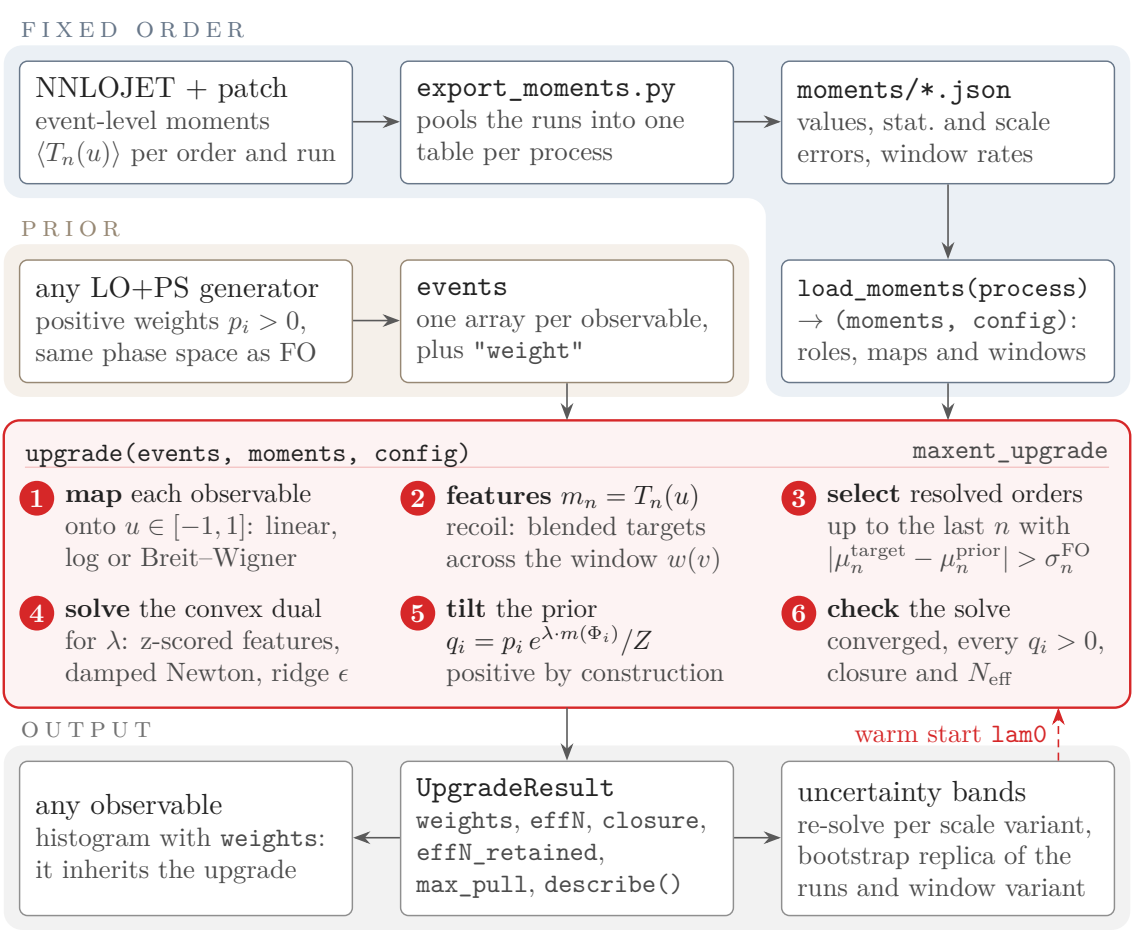

FIXED ORDER
NNLOJET + patch
event-level moments
$\langle T_n(u) \rangle$ per order and run
export_moments.py
pools the runs into one table per process
moments/*.json
values, stat. and scale errors, window rates
PRIOR
any LO+PS generator
positive weights $p_i > 0$, same phase space as FO
events
one array per observable, plus "weight"
load_moments(process)
→ (moments, config):
roles, maps and windows
upgrade(events, moments, config)
maxent_upgrade
1 map each observable onto $u \in [-1, 1]$: linear, log or Breit–Wigner
2 features $m_n = T_n(u)$ recoil: blended targets across the window $w(v)$
3 select resolved orders up to the last n with $|\mu_n^{\text{target}} - \mu_n^{\text{prior}}| > \sigma_n^{\text{FO}}$
4 solve the convex dual for λ: z-scored features, damped Newton, ridge ϵ
5 tilt the prior $q_i = p_i\, e^{\lambda \cdot m(\Phi_i)}/Z$ positive by construction
6 check the solve converged, every $q_i > 0$, closure and N_{eff}
OUTPUT
warm start lam0
any observable
histogram with weights: it inherits the upgrade
UpgradeResult
weights, effN, closure, effN_retained, max_pull, describe()
uncertainty bands
re-solve per scale variant, bootstrap replica of the runs and window variant